

Statement of Work

3D-TLS Scanner 2 Year Maintenance

Upon receipt of scanner and acceptance in the Government Furnished Property (GFP) module of WAWF:

- Complete 2 year System Maintenance.
- Basic Service Package A
- Readjust the range measurement unit of the instrument
- Update firmware -> can only be done if extended software maintenance is purchased
- Test in the climate chamber and readjust at different temperatures
- Recalibrate the internal calibration at the "Riegl-Test-Field"
- Recalibrate the internal orientation sensors (imu, gyro, inclination)
- Recalibrate the internal compass (magnetic field sensor)
- Check all interfaces (LAN 100MB, 1GB, WLAN, GPS, USB Camera, USB-Stick)
- Check the current consumption of the scanner unit
- End test of electrical performance (max. range, range accuracy, etc.)
- Check the parameters of the instrument
- Modification of the control pcb
- Check the Add-On Rechargeable Battery RBNE 2210 (01219)
- Replace the desiccant cartridge and reflush with dry nitrogen
- Prepare the service protocol and test certificate
- Clean and return

Scanners will meet the following Riegl 3D-TLS required specifications upon completion of service and calibration.

Laser Pulse Repetition Rate PRR (peak)	50 kHz	100 kHz	300 kHz	600 kHz	1.2 MHz
Effective Measurement Rate (meas./sec)	21,000	42,000	125,000	250,000	500,000
Max. Measurement Range	2,500 m	1,850 m	1,100 m	800 m	600 m
blank	1,300 m	950 m	540 m	380 m	290 m
Max. Number of Targets per Pulse	Up to 15	up to 15	Up to 15	Up to 8	Up to 4
Accuracy	5 mm	5 mm	5 mm	5 mm	5 mm
Precision	3 mm	3 mm	3 mm	3 mm	3 mm
Minimum Range	2.0 m	1.5 m	1.5 m	1.0 m	1.0 m
Effective Measurement Rate (meas./sec)	21,000	42,000	125,000	250,000	500,000
Laser Wavelength	Near Infrared				
Beam Divergence	0.27 mrad				

Scanners will be returned to Hill AFB, UT using 84 RADES shipping account and transferred back to RADES in the WAWF GFP module.